SHEAFIFYING THE LEFT KAN EXTENSION BEFORE OR AFTER SHEAFIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The direct image of a sheaf along a continuous functor. Same chapter, Lemma lemma-pushforward-sheaf, whose situation the statement refers to. Supplies the identification of the direct image on sheaves and the adjunction with the left Kan extension.
- The Yoneda lemma. Categories, standard form of the Yoneda lemma for the category of sheaves. Turns the displayed chain of bijections into an isomorphism of sheaves.

Source reference. Original lemma 00WW: sources/stacks-sites/pinned-original-source.tex, source lines 2797–2797. Navigation pointer invoked at proof line 2797; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *In the situation of Lemma 00WW. For any presheaf $\mathcal{G}$ on $\mathcal{C}$ we have $(u_p\mathcal{G})^\# = (u_p(\mathcal{G}^\#))^\#$.*

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. For any sheaf $\mathcal{F}$ on $\mathcal{D}$ we have

$$\begin{aligned} \mathrm{Mor}_{Sh(\mathcal{D})}(u_s(\mathcal{G}^\#), \mathcal{F}) &= \mathrm{Mor}_{Sh(\mathcal{C})}(\mathcal{G}^\#, u^s\mathcal{F}) \\ &= \mathrm{Mor}_{PSh(\mathcal{C})}(\mathcal{G}^\#, u^p\mathcal{F}) \\ &= \mathrm{Mor}_{PSh(\mathcal{C})}(\mathcal{G}, u^p\mathcal{F}) \\ &= \mathrm{Mor}_{PSh(\mathcal{D})}(u_p\mathcal{G}, \mathcal{F}) \\ &= \mathrm{Mor}_{Sh(\mathcal{D})}((u_p\mathcal{G})^\#, \mathcal{F}) \end{aligned}$$

and the result follows from the Yoneda lemma. □